
SATURN: Splitting Approach on Tensorized Ubiquitous Regression Networks

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Abstract

1 Convolutional Neural Networks often contain fully connected layers after their
2 convolution phase, which requires the data to be flattened. Despite empirical
3 success, this flattening ruins the multidimensional structure of the input data and
4 increases the number of parameters, which may cause overfitting. Given the tensor
5 nature of the input data, fully connected layers can be replaced with tensorized low-
6 rank layers, which, besides considering the nature of the data, reduce the number
7 of parameters. However, these layers perform poorly when the approximated rank
8 is too low. To address this issue and further reduce the number of parameters
9 while preserving the quality of the network, we propose a framework that, by
10 using two splitting methods, transforms the multidimensional tensor into a thinner
11 and higher-order tensor and applies tensorized layers on this high-order tensor.
12 This leads to fewer parameters and less computational cost while preserving the
13 quality of the network. In our experiments, evaluated on CIFAR10, MNIST, and
14 Tiny ImageNet-200, we have successfully reduced the classifier parameters of
15 well-known models such as VGG19, ResNet50, and ResNet101 by more than 95%,
16 while also achieving better accuracy and overcoming overfitting. Compared to
17 traditional state-of-the-art models, our approach achieves significant parameter
18 reduction and compression while preserving accuracy.

19 1 Introduction

20 Neural Networks, especially those that are applied to images such as *Convolutional Neural Network*
21 (CNNs), often consist of two main parts: One that is responsible to extract features from the input
22 data, and the other part is a *MultiLayer Perceptron* (MLP) which Maps the extracted features to
23 the desired space. Despite the importance of data and their spatial relationship, maintaining their
24 topological structure is often discarded since many known CNNs like ResNet or VGG, seem to work
25 fine; However in cases where these structures are more important than the others such as MRI, It
26 is crucial to adapt the known networks to work without destroying the multi-linear structure of the
27 data. While preserving the natural structure of the data throughout the network is important, a greater
28 drawback from using MLPs is their excessive required number of parameters which not only limit
29 their capabilities in real time and practical problems but also prone them to overfitting.

30 Most natural datasets can be represented as a Tensor due their multi-linear or multi-modal structure;
31 Images can be considered as a 3-order tensors with width, height and the color channels as their
32 corresponding 3 modes, the same could be considered for videos as 4-order tensor with an extra time
33 mode. In practice, most of the used data in the current machine learning and artificial intelligence
34 tasks can be encoded as tensors, requiring their expansion to the linear algebra.

35 In machine learning, we usually consider data as vectors and datasets will form a set of vectors
36 (ideally called a matrix). By that definition, many matrix decomposition methods have been invented

to work on such matrices. Tensor methods are to generalize these methods to a higher dimensional space. Mathematically, tensors, have been widely studied in the past. One excellent example within tensor methods is Tensor Decomposition which can be considered as a prime backbone for most tensorized layers and approximations in deep learning [1] [2], Signal Processing [3] [4], fusion [5], blind source separation [4] [6], and computer vision [7] .

Considering the new advances in *Deep Neural Networks* (DNNs) and their applications to computer vision such as image recognition, image segmentation, image generation, video generation, vision transformers, etc . . . ; most architectures still flatten the data before feeding them to their MLPs which not only destroys the natural structure of the data but also increase the number of parameters needed for the model.

Images as the frequent example of data used in machine learning tasks are considered 3-order tensors which require their own algebra to be applied when fed to the MLP. *Fully Connected* (FC) layers used in many architectures are the main culprit behind vectorizing the data and the number of parameters. To alleviate this situation, a better replacement for FC layers is needed which not only preserves the current model’s accuracy but also reduces the number of required parameters and does not destroy the structure of data.

One possible working solution to tensorizing FC layers is by employing the low rank approximation of the weight tensor. Some works have been done to address these issues which will be discussed in Section 2 . However, those methods usually perform poorly when the approximated rank is too low which is not ideal for larger models. The cost of these models usually depend on the size of the tensor, our experiment shows that a thinner while higher dimensional reshaped tensor of the original input tensor, can perform much better in lower ranks and by a unique splitting and stacking technique, we preserved the models’ accuracy. In this paper, we have provided two splitting approaches which have their own use dependent on the model’s task; Using these methods, we have reshaped the tensor to a higher dimensional space where the cost of computation is less and has a better capability to handle lower approximated ranks. Our methods evaluated on CIFAR10, MNIST, and Tiny-ImageNet-200 applied to known CNN models such as ResNet50, ResNet101, and VGG19, as well as a typical CNN achieved similar accuracy in most methods with more than 95 percent less parameters compared to the FC layers and even overcame overfitting in some cases.

2 Related works

Several prior papers address the importance of using tensors and preserving the natural structure, One uses CP decomposition to enhance convolution layers [8]. In [9], tucker decomposition is used for convolution enhancement with fewer parameters. A weight sharing scheme in multi tasking learning is proposed by [10]. In [11], a Tensor-Train decomposition is used as a low rank approximation of the weight tensor. Some uses tensor methods for model compression and acceleration [12].

In general, tensor methods can be widely used in many machine learning and deep learning tasks, especially computer vision [13]. Their use, can be categorized into three main approaches: The first approach compresses the tuned parameters of the pre-trained network while preserving the quality of network with these approximated parameters. So here the new network is not trained and the approximated network could only be used in the test step. Two methods for approximation of the tuned parameters of pertained CNN in each layer based on matrix and tensor decompositions adjusted with clustering of inlet and outlet layers have been proposed [14]. An improvement is reported in [15]. Tensor CP and Tucker decompositions are used to approximate the kernels of pre-trained CNN networks [16] [17] [18]. In [18] a tensor based method to find an approximation of fully connected layer parameters in CNN networks is introduced. Despite the proper implementation results, these methods have the following drawbacks: increase cost of pre-trained networks and finding these compressions is also costly. It also cannot be used in applications such as deep reinforcement learning where the network constantly should be learned. On the other hand, this approach has confined itself to compressing convolutional methods for the test phase, although a deeper question can be asked. Why should we compress an already trained network and not seek for a compact network representation that can be trained from scratch?

The second approach consider the convolutional layers as a base, and approximate or represent them with smaller parameters, to design end-to-end trainable neural network. Different kinds of methods have been proposed to design end-end learnable networks based on approximation of convolution

process with different views: For example, in [19] and [20], the rank-1 approximation of 3-order tensor filters corresponding to each outlet channel is used. In [21], share factors corresponding to inlet and outlet are considered to reduce the redundancy of parameters. Recently, in [22] by considering the 3-order filters of each outlet channel as rank-k tensor it has been shown that the pixels of each outlet channel is the contraction of this tensor with the corresponding patch of all input channels. Unlike standard convolution, Mobile Nets proposed in [23] separate the spatial filtering and integration steps. For each input channel, only a single spatial filtering is performed, and by different combination of these filtered images, different output channels are obtained.

The third approach do not consider Convolutional structure as a base and try to find other type of layers to work on multidimensional data. One notable recent work is Tensor Regression Networks [1]. They used a *Tensor Regression Layer* (TRL) and *Tensor Contraction Layer* (TCL) to preserve the structure of the data as they are moving across the network while reducing the number of parameters. Their method surprisingly achieved the near similar accuracy as its FC network with more than 65 percent space saving. TRL and TCL will be discussed in section 3, and they are used as the base of our method. Unfortunately, one of the biggest issues with TCL and TRL is their inadequate implementations. They are theoretically required less parameters than their FC versions; However they often perform poorly in terms of the actual execution time. This is one of the biggest limitation on working with tensors since a practical implementation of tensor methods are yet to be made.

The power of tensor regression has been addressed in many prior papers [24][25] [26] [27]. However, most of these works, usually rely on analytical solution and large tensors. In [28], a non-linear Tensor-Train decomposition is used instead of tucker decomposition. While many works focuses on individual layers, in [29], T-Nets attempts to provide a single high-order, low-rank tensor that jointly capture the full structure of a neural network.

DNNs have been used in many applications and problems, however, many questions still remains in terms of their credibility and why they work so well. DNNs usually contain too many parameters and an important question is whether they require that many parameters. By tensorizing the DNNs, we can address these issues and attempt to better understand the concepts of deep learning. For example, one uses tensor methods as a tool to study CNNs [30], they follow up by studying the power of overlapping architectures in deep learning [31]. Some provide global optimal and optimization for non-convex factorization problems [32]. In [33], they used contraction as a composition operator in *Natural Language Processing* (NLP). Some used tensor methods as a tool to guarantee convergence and consistent learning[34].

Most of the recent works, rely on low rank approximation of the tensor or some tensor decomposition methods as a dimension reduction layer. Despite their success, a fully tensorized end-to-end trainable model requires almost near similar ranks to perform well when compared to their FC competitives. The draw back of using same ranks as the input tensor is the cost. Our method provide a reshaping approach that maintains the accuracy by converting the input tensor into a thinner and higher order tensor that requires fewer parameters and can perform better in lower ranks. To our knowledge, no prior work increases the dimensions of the tensor in order to achieve near similar accuracy while using tensorized layers with lower ranks and fewer parameters.

3 Mathematical background

This paper, requires basic understanding of tensor methods, If readers are already knowledgeable in the matter, they can skip to section 4.

3.1 Notations

For a 3-order tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, there are 3 modes with respected sizes I_1 , I_2 , and I_3 . Its $\mathcal{T}_{i_1, i_2, i_3}$ **element** is a 0-order tensor (scalar), (i_1, i_2, i_3) . **Slices** of its first and second modes are 2-order tensors (matrices) $\mathbf{M}$ and are denoted by two colons, $\mathcal{T}_{:, :, i_3}$. **Fibers** of its first mode are 1-order tensors (vectors) $\mathbf{V}$ and are denoted by a colon, $\mathcal{T}_{:, i_2, i_3}$.

139 3.2 Tensor folding and unfolding

140 Tensor folding and unfolding are fundamental operations in tensor analysis, allowing for the manipula-
 141 tion and transformation of tensors into different dimensional structures. Tensor unfolding, also known
 142 as matricization, converts a higher-order tensor into a matrix or a vector. Let $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ be
 143 an N -order tensor. In our case, tensor folding is the inverse operation of tensor unfolding, which
 144 reconstructs a higher-order tensor from a matrix or vector.

145 **Mode- n unfolding :** Unfolding $\mathcal{T}$ along mode- n [2] results in a matrix representation of mode- n .
 146 This operation is denoted as $T_{[n]}$. The unfolding is achieved by rearranging the entries of $\mathcal{T}$ into a
 147 matrix such that the mode- n indices become the rows of the matrix. The mapping from tensor indices
 148 to matrix indices is given by:

$$\begin{aligned} (i_1, \dots, i_n, \dots, i_N) &\longrightarrow (i_n, r) \\ \text{where } r &= \sum_{\substack{p=1 \\ p \neq n}}^N i_p \times \prod_{\substack{l=p+1 \\ l \neq n}}^N I_l \end{aligned} \quad (1)$$

149 Each row of the resulting matrix corresponds to a different combination of indices for the remaining
 150 modes.

151 **Vectorization :** Similar to mode- n unfolding, in tensor vectorization [2] we map each element
 152 within the tensor to a element j of $Vec(\mathcal{T})$:

$$\begin{aligned} (i_1, \dots, i_n, \dots, i_N) &\longrightarrow (r) \\ \text{where } r &= \sum_{p=1}^N i_p \times \prod_{l=p+1}^N I_l \end{aligned} \quad (2)$$

153 3.3 Tensor contraction

154 Tensor contraction allows for the multiplication and simplification of tensors and provides a way to
 155 combine tensors by summing over shared indices.

156 **Tensor n -mode product :** Tensor n -mode product [2], is a tensor operation that allows for the
 157 multiplication of a tensor with a matrix along a specific mode. Given a tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$
 158 and a matrix $\mathbf{M} \in \mathbb{R}^{J \times I_n}$, the n -mode product of $\mathcal{T}$ and $\mathbf{M}$, denoted as $\mathcal{T} \times_n \mathbf{M}$, results in a tensor
 159 of size $\mathbb{R}^{I_1 \times \dots \times I_{n-1} \times J \times I_{n+1} \times \dots \times I_N}$. The n -mode product can be expressed mathematically as:

$$(\mathcal{T} \times_n \mathbf{M})_{i_1, \dots, i_{n-1}, j, i_{n+1}, \dots, i_N} = \sum_{i_n=0}^{I_n} \mathcal{T}_{i_1, \dots, i_n, \dots, i_N} \mathbf{M}_{j, i_n} \quad (3)$$

160 where $i_1, \dots, i_{n-1}, j, i_{n+1}, \dots, i_N$ are the indices of the resulting tensor, and i_n is the summation
 161 index. Alternatively, the n -mode product can be expressed using the tensor unfolding operation along
 162 the n -th mode, denoted as $\mathbf{T}_{(n)}$. In this case, the n -mode product can be written as:

$$\mathcal{T} \times_n \mathbf{M} = \mathbf{M} \mathbf{T}_{(n)} \quad (4)$$

163 **Generalized inner product :** For two tensors $\mathcal{X} \in \mathbb{R}^{I_x \times I_1 \times I_2 \times \dots \times I_N}$ and $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times I_y}$
 164 sharing N modes of the same size, a generalized inner product [2] along the N last modes of X and
 165 Y is defined as:

$$\langle \mathcal{X}, \mathcal{Y} \rangle_N = \sum_{i_1=0}^{I_1-1} \dots \sum_{i_N=0}^{I_N-1} \mathcal{X}_{:, i_1, i_2, \dots, i_N} \mathcal{Y}_{i_1, i_2, \dots, i_N, :} \quad (5)$$

166 where $\langle \mathcal{X}, \mathcal{Y} \rangle_N \in \mathbb{R}^{I_x \times I_y}$.

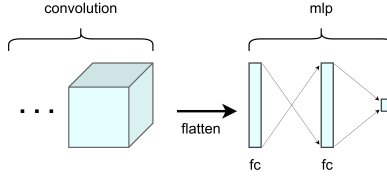


Figure 1: A simple MLP after some convolution layers.

3.4 Tucker decomposition

The Tucker decomposition [2] is a powerful method in tensor algebra that breaks down a tensor into a core tensor multiplied by matrices along each mode. In the Tucker decomposition, a tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ is represented as the product of a core tensor $\mathcal{G} \in \mathbb{R}^{J_1 \times J_2 \times \dots \times J_N}$ and a set of factor matrices $\mathbf{A}^{(1)} \in \mathbb{R}^{I_1 \times J_1}$, $\mathbf{A}^{(2)} \in \mathbb{R}^{I_2 \times J_2}$, $\dots$, $\mathbf{A}^{(N)} \in \mathbb{R}^{I_N \times J_N}$ along each mode. Mathematically, the Tucker decomposition can be expressed as:

$$\mathcal{T} = \mathcal{G} \times_1 \mathbf{A}^{(1)} \times_2 \mathbf{A}^{(2)} \times \dots \times_N \mathbf{A}^{(N)} \quad (6)$$

where $\times_n$ denotes the n -mode product. The Tucker decomposition can be computed using the mode- n unfolding of the tensor, denoted as $\mathbf{T}_{[n]}$. By reshaping the tensor along each mode, the Tucker decomposition can be written as:

$$\mathbf{T}_{[n]} = \mathbf{A}^{(n)} \mathbf{G}[n] \left(\mathbf{A}^{(1)} \otimes \dots \otimes \mathbf{A}^{(n-1)} \otimes \mathbf{A}^{(n+1)} \otimes \dots \otimes \mathbf{A}^{(N)} \right)^T \quad (7)$$

where $\mathbf{G}[n]$ represents the unfolding of the core tensor along mode n .

3.5 Tensor regression networks

The combination of tensor regression and tensor contractions has been used in an end-to-end trainable model by Kossaifi [1], as discussed in section 2. Given a MLP, there are two ways to interpret FC layers: One is to consider them as a feature extraction layer where they map a latent or feature space to another latent space, on another hand, we can consider them as regression layers which are basically the classifier layers and given the purpose of the network, they can be used interchangeably. In figure 1 a very simple MLP that is being used after some convolution layers is provided. In this case, the first FC layer can be considered a dimension reduction layer while the second FC layer maps the feature space to the target space. In the following text, we will discuss to main approaches to replace such FC layers:

3.5.1 Tensor contraction layer

Tensor Contraction Layer (TCL) [1] is a form of feature extraction and could also be used for embedding. The prime idea behind TCL is Tucker decomposition; Via multiplying matrices to each mode of the input tensor, we can obtain a denser tensor (dependent of the shape of the matrices). Figure 2 demonstrate the TCL process on a 3-order tensor.

Considering an input tensor $\mathcal{X}$ of size $(S_0, I_1, \dots, I_N)$ and a size $(R_1, \dots, R_N)$ TCL, where S_0 is the size of the batch and (I_i, R_i) is the shape of the matrix applied to mode i of the tensor; The number of required parameters is calculated as :

$$\sum_{i=1}^N R_i \times I_i \quad (8)$$

And the resulting tensor is of size $(S_0, R_1, \dots, R_N)$. Compared to the FC version where the simple regression between these two tensors would require at least $\prod_{i=1}^N I_i \times R_i$ parameters.

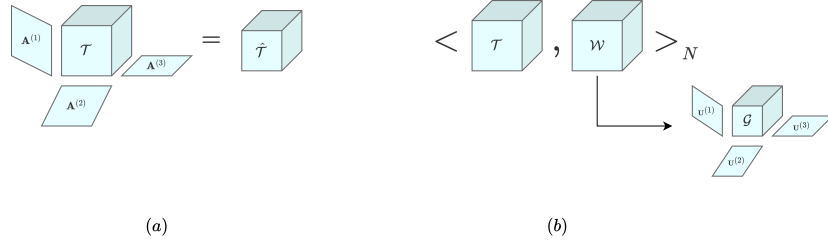


Figure 2: Tensor regression networks, replacement for FC layers. In (a), A simple TCL is applied; Each mode i of tensor $\mathcal{T}$ is multiplied by factor $\mathbf{A}^{(i)}$. In (b), A TRL is applied; Generalized inner product of input tensor $\mathcal{T}$ and weight tensor $\mathcal{W}$ is calculated as the output tensor while considering a low rank approximation of weight tensor $\mathcal{W}$ in a tucker decomposed form.

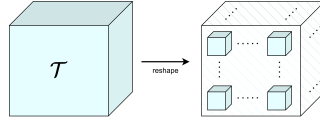


Figure 3: An outline of reshaping to higher orders tensor.

197 3.5.2 Tensor regression layer

198 *Tensor Regression Layers* (TRL) [1] is a replacement for the FC regression layers. A simple regression
 199 that is the foundation of a MLP is a trivial generalized inner product of $\mathcal{W}$ (Weight Tensor) with
 200 the $\mathcal{X}$ (Input tensor). The idea behind TRL is to reconstruct $\mathcal{W}$ from a low rank approximation of
 201 itself via tucker decomposition. In figure 2, a TRL application in a MLP is provided. For the sake of
 202 simplicity, we considered a binary classification problem where the output of the inner product is a
 203 scalar. Similarly, if the desired output is a vector or even a tensor, $\mathcal{W}$ will be a higher dimensional
 204 tensor itself to generate each mode of the output.

205 Considering an input tensor $\mathcal{X}$ of size $(S_0, I_1, \dots, I_N)$, an output tensor $\mathcal{Y}$ of size $S_0 \times O$, and a
 206 size $(R_1, \dots, R_N, R_{N+1})$ TRL with bias b of size O such that $\mathcal{Y} = \langle \mathcal{X}, \mathcal{W} \rangle_N + b$; The number of
 207 required parameters can be calculated as :

$$\prod_{i=1}^{N+1} R_i + \left(\sum_{i=1}^N R_i \times I_i + R_{N+1} \times O \right) \quad (9)$$

208 Compared to the FC version where the number of parameters is $O \times \prod_{i=1}^N I_i$.

209 4 Splitting approach

210 Impressed by the vision transformers patching and down sampling methods, we can transform
 211 our input tensor into a thinner and higher dimensional tensor that not only reduces the number of
 212 parameters for a TRL or TCL layer but also maintain the accuracy. However, not any reshaping
 213 scheme to arbitrary higher orders and dimensions guarantee that the multi-linear structure is preserved
 214 and the accuracy is maintained. With a proper splitting scheme, the newly made tensor has better
 215 physical interpretation than the flattened data. We introduced two splitting schemes, both of which
 216 are useful and could be used differently according to the network architect. Figure 3 demonstrate the
 217 general splitting outline. Given a 3-order tensor, if we split across each mode, the output is a 6-order
 218 tensor where the first 3 modes correspond the split index, and the next 3 modes are the split data.

219 4.1 Attached splitting

220 In attached splitting, every split is continues, that is if mode I_i of the input tensor $\mathcal{X}$ with length n is
 221 to be split into 2 parts, then the split indices are $(0, \frac{n}{2})$ and $(\frac{n}{2}, n)$. Figure 4 demonstrate the splitting
 222 process on a matrix.

223 Given a tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ and split sizes $(L_1, \dots, L_N)$, attached mapping of elements is done
 224 via:

$$\mathcal{T}[i_1, \dots, i_N] \longrightarrow \mathcal{T}_a \left[\lfloor \frac{L_1 \times i_1}{I_1} \rfloor, \dots, \lfloor \frac{L_N \times i_N}{I_N} \rfloor, i_1 \bmod \frac{I_1}{L_1}, \dots, i_N \bmod \frac{I_N}{L_N} \right] \quad (10)$$

225 where $\mathcal{T}_a \in \mathbb{R}^{L_1 \times \dots \times L_N \times \frac{I_1}{L_1} \times \dots \times \frac{I_N}{L_N}}$.

226 4.2 Compressed splitting

227 In compressed splitting, we incorporate the idea behind down sampling, that is if mode I_i of input
 228 tensor $\mathcal{X}$ with length n is to be split into 2 part, different to the attached splitting, the indices are
 229 even-odd combinations; That is $\{t|t \in (0, n) \wedge (t \bmod 2 = 0)\}$ and $\{t|t \in (0, n) \wedge (t \bmod 2 = 1)\}$.

230 Given a tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ and split sizes $(L_1, \dots, L_N)$, compressed mapping of elements is
 231 done via:

$$\mathcal{T}[i_1, \dots, i_N] \longrightarrow \mathcal{T}_c \left[i_1 \bmod L_1, \dots, i_N \bmod L_N, \lfloor \frac{i_1}{L_1} \rfloor, \dots, \lfloor \frac{i_N}{L_N} \rfloor \right] \quad (11)$$

232 where $\mathcal{T}_c \in \mathbb{R}^{L_1 \times \dots \times L_N \times \frac{I_1}{L_1} \times \dots \times \frac{I_N}{L_N}}$.

233 4.3 Method analysis

234 Considering an input tensor $\mathcal{X}$ of size $(S_0, I_1, \dots, I_N)$, either reshaping would result in the tensor
 235 $\hat{\mathcal{X}}$ of size $(S_0, L_1, \dots, L_N, \frac{I_1}{L_1}, \dots, \frac{I_N}{L_N})$. An example of each reshaping approach is done in figure
 236 5 on a sample image.

237 In case of a $(r_1, \dots, r_N, R_i, \dots, R_N)$ TCL, Since L_i s are usually small, the number of parameters
 238 required for them $\sum_{i=1}^N L_i \times r_i$ is also small; Similarly, $\frac{I_i}{L_i}$ is smaller than I_i so the new desired R_i s
 239 is smaller than the previous R_i s and the total number of parameters can be calculated as:

$$\sum_{i=1}^N L_i \times r_i + \sum_{i=1}^N R_i \times \frac{I_i}{L_i} \quad (12)$$

240 which is ideally less than equation 8 (Appendix A).

241 In case of $(r_1, \dots, r_N, R_1, \dots, R_N, R_{N+1})$ TRL, the same logic as TCL could also be applied. The
 242 total number of parameters in a new TRL is calculated as:

$$\left(\prod_{i=1}^N r_i \prod_{i=1}^{N+1} R_i \right) + \left(\sum_{i=1}^N L_i \times r_i + \sum_{i=1}^N R_i \times \frac{I_i}{L_i} + R_{N+1} \times O \right) \quad (13)$$

243 which is ideally less than equation 9 (Appendix B).

244 5 Experiments

245 6 Conclusion

246 Deep neural networks, especially CNNs, often work with multi-dimensional tensors. Despite their
 247 empirical success, these architectures usually contain one or more fully connected layers which

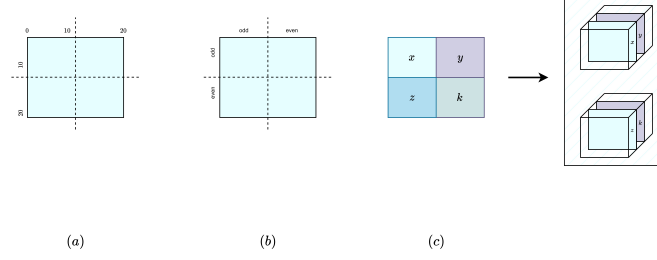


Figure 4: (a) attached splitting of a 2-order tensor. (b) compressed splitting of a 2-order tensor. (c) demonstration of stacking split chunks of a 2-order tensor into a 4-order tensor.

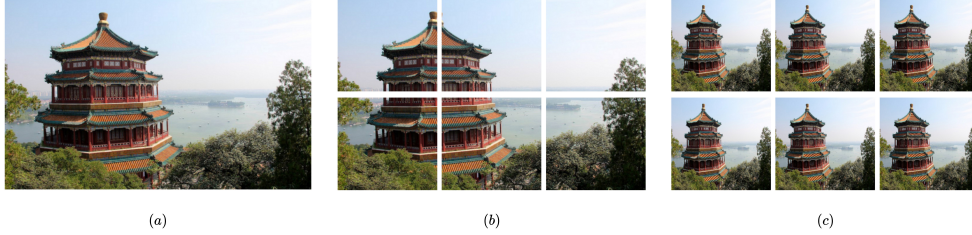


Figure 5: (a) original sample image. (b) reshaped sample image using attached approach. (c) reshaped sample image using compressed approach.

not only destroy the natural multi-linear structure of the data, but also over parameterize the model. By introducing tensorized layers such as TRL and TCL we can achieve near similar accuracy while preserving the multi-modal structure of the data and reducing the number of classifier parameters; However, these methods perform poorly when the approximated rank is too low. We introduced SATURN, a framework which maintains the quality of the network and reduce the number of classifier parameter by more than 96 %. Our experiment, evaluated on CIFAR10, MNIST, and Tiny ImageNet-200, applied to RestNet50, RestNet101, VGG19 , and classic CNNs demonstrate that by splitting the input tensor via an approach that does not disturb the multi-linear structure of the data, a TCL or TRL with fewer parameters can be performed which also maintains the accuracy. Based on our observations, many popular CNNs learn their convolutional layers based on the flattening operation and their following FC layers. Therefore, for our framework to achieve best accuracy, the models need to be trained completely for a few iterations. While our framework and similar approaches offer theoretical advantages, they currently do not exhibit the expected speed. This limitation underscores the primary challenge in applying tensor methods to models, as the development of optimal implementations remains a pending task.

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A Method analysis on TCL

As stated in subsection 4.3, equation 12, the number of required parameters for a TCL on our framework is ideally less than equation 8. Given $\mathcal{T} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ and $\hat{\mathcal{T}} \in \mathbb{R}^{L_1 \times \dots \times L_N \times \frac{I_1}{L_1} \times \dots \times \frac{I_N}{L_N}}$, ideally a rank $(r_1, \dots, r_N, \hat{R}_1, \dots, \hat{R}_N)$ TCL on $\hat{\mathcal{T}}$ has less parameters than a rank $(R_1, \dots, R_N)$ TCL on $\mathcal{T}$. Since TCL ranks are arbitrary values, one can choose these ranks in a way that inequality 14 does not hold.

$$\sum_{i=1}^N L_i \times r_i + \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} \leq \sum_{i=1}^N R_i \times I_i \quad (14)$$

ideal. We prove that given ideal ranks, inequality 14 holds.

Assume $\forall i \ \hat{R}_i \leq R_i$, then:

$$\sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} \leq \sum_{i=1}^N R_i \times I_i \quad \forall i \ L_i \geq 1 \quad (15)$$

Consider α as:

$$\begin{aligned}
\alpha &= \sum_{i=1}^N R_i \times I_i - \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} \\
&= \sum_{i=1}^N (R_i - \frac{\hat{R}_i}{L_i}) I_i
\end{aligned} \tag{16}$$

358 Then as long as the first part of equation 12 is less than α , the inequality 14 holds.

$$\sum_{i=0}^N r_i \times L_i \leq \alpha \tag{17}$$

359 Given the assumption and the inequality 17:

$$\begin{aligned}
\sum_{i=1}^N L_i \times r_i + \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} &\leq \alpha + \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} \\
&\leq \sum_{i=1}^N R_i \times I_i - \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} + \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} \\
&\leq \sum_{i=1}^N R_i \times I_i
\end{aligned} \tag{18}$$

360 Therefore, with ideal ranks, $\forall i \ \hat{R}_i \leq R_i$ and $\sum_{i=0}^N r_i \times L_i \leq \alpha$, inequality 14 holds. $\square$

361 **semi ideal.** We prove that given semi ideal ranks, inequality 14 holds.

362 Assume $\forall i \ \hat{R}_i > R_i$, then:

$$k_i = \hat{R}_i - R_i \quad \forall i \tag{19}$$

363 We can consider:

$$\sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} = \sum_{i=1}^N R_i \times \frac{I_i}{L_i} + \sum_{i=1}^N k_i \times \frac{I_i}{L_i} \tag{20}$$

364 Consider β as:

$$\begin{aligned}
\beta &= \sum_{i=1}^N R_i \times I_i - \sum_{i=1}^N R_i \times \frac{I_i}{L_i} \\
&= \sum_{i=1}^N R_i \times I_i \times (1 - \frac{1}{L_i})
\end{aligned} \tag{21}$$

365 Then as long as the second part of equations 20 is less than β , the inequality 15 holds.

$$\sum_{i=1}^N k_i \times \frac{I_i}{L_i} \leq \beta \tag{22}$$

366 Given the assumption and the inequality 22:

$$\begin{aligned}
\sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} &= \sum_{i=1}^N R_i \times \frac{I_i}{L_i} + \sum_{i=1}^N k_i \times \frac{I_i}{L_i} \\
&\leq \sum_{i=1}^N R_i \times \frac{I_i}{L_i} + \beta \\
&\leq \sum_{i=1}^N R_i \times \frac{I_i}{L_i} + \sum_{i=1}^N R_i \times I_i - \sum_{i=1}^N R_i \times \frac{I_i}{L_i} \\
&\leq \sum_{i=1}^N R_i \times I_i
\end{aligned} \tag{23}$$

367 Therefore, with ideal splits, $\sum_{i=1}^N k_i \times \frac{I_i}{L_i} \leq \beta$, inequality 15 holds. The rest of the proof is similar
368 to *ideal* situation. $\square$

369 **semi ideal.** We prove that given some ideal ranks, inequality 14 holds.

370 Assume $\exists j \ \hat{R}_j > R_j$, then:

$$k_j = \hat{R}_j - R_j \quad \forall j \text{ s.t. } \hat{R}_j > R_j \tag{24}$$

371 We can consider $N_1 = \{ \forall i \mid \hat{R}_i \leq i \}$ and $N_2 = \{ \forall j \mid \hat{R}_j > j \}$:

$$\begin{aligned}
\sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} &= \sum_{i \in N_1} \hat{R}_i \times \frac{I_i}{L_i} + \sum_{j \in N_2} \hat{R}_j \times \frac{I_j}{L_j} \\
&\leq \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} \hat{R}_j \times \frac{I_j}{L_j} \\
&\leq \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times \frac{I_j}{L_j} + \sum_{j \in N_2} k_j \times \frac{I_j}{L_j}
\end{aligned} \tag{25}$$

372 Consider β as:

$$\begin{aligned}
\beta &= \sum_{i=1}^N R_i \times I_i - \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times \frac{I_j}{L_j} \\
&= \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times I_j - \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times \frac{I_j}{L_j} \\
&= \sum_{j \in N_2} R_j \times I_j - \sum_{j \in N_2} R_j \times \frac{I_j}{L_j} \\
&= \sum_{j \in N_2} R_j \times I_j \times (1 - \frac{1}{L_j})
\end{aligned} \tag{26}$$

373 Then as long as the last part of inequality 25 is less than β , the inequality 15 holds.

$$\sum_{j \in N_2} k_j \times \frac{I_j}{L_j} \leq \beta \tag{27}$$

374 Given the assumption and the inequality 27:

$$\begin{aligned}
\sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} &\leq \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times \frac{I_j}{L_j} + \sum_{j \in N_2} k_j \times \frac{I_j}{L_j} \\
&\leq \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times \frac{I_j}{L_j} + \beta \\
&\leq \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times \frac{I_j}{L_j} + \sum_{j \in N_2} R_j \times I_j - \sum_{j \in N_2} R_j \times \frac{I_i}{L_i} \\
&\leq \sum_{i \in N_1} R_i \times I_i + \sum_{j \in N_2} R_j \times I_j \\
&\leq \sum_{i=1}^N R_i \times I_i
\end{aligned} \tag{28}$$

375 Therefore, with ideal splits as long as the existed semi ideal ranks follow inequality 27, inequality 15
376 holds. The rest of the proof is similar to *ideal* situation. $\square$

377 B Method analysis on TRL

378 As stated in subsection 4.3, equation 13, the number of required parameters for a TRL on our frame-
379 work is ideally less than equations 9. Given $\mathcal{T} \in \mathbb{R}^{S_0, I_1 \times \dots \times I_N}$ and $\hat{\mathcal{T}} \in \mathbb{R}^{S_0, L_1 \times \dots \times L_N \times \frac{I_1}{L_1} \times \dots \times \frac{I_N}{L_N}}$,
380 and output $\mathcal{Y} \in \mathbb{R}^{S_0 \times O}$, ideally a rank $(r_1, \dots, r_N, \hat{R}_1, \dots, \hat{R}_N, \hat{R}_{N+1})$ TRL on $\hat{\mathcal{T}}$ has less param-
381 eters than a rank $(R_1, \dots, R_N, R_{N+1})$ TRL on $\mathcal{T}$. Since TRL ranks are arbitrary values, one can
382 choose these ranks in a way that inequality 29 does not hold.

$$\prod_{i=1}^N r_i \prod_{i=1}^{N+1} \hat{R}_i + \sum_{i=1}^N L_i \times r_i + \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \leq \prod_{i=1}^{N+1} R_i + \sum_{i=1}^N R_i \times I_i + R_{N+1} \times O \tag{29}$$

383 *ideal*. We prove that given ideal ranks, inequality 29 holds.

384 Assume $\forall i \ \hat{R}_i \leq R_i$, then:

$$\sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \leq \sum_{i=1}^N R_i \times I_i + R_{N+1} \times O \quad \forall i \ L_i \geq 1 \tag{30}$$

385 Consider α_1 as:

$$\begin{aligned}
\alpha_1 &= \sum_{i=1}^N R_i \times I_i + R_{N+1} \times O - \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \\
&= \sum_{i=1}^N (R_i - \frac{\hat{R}_i}{L_i}) \times I_i + (R_{N+1} - \hat{R}_{N+1}) \times O
\end{aligned} \tag{31}$$

386 Then as long as the second part of equation 13 is less than α_1 ,

$$\sum_{i=1}^N L_i \times r_i \leq \alpha_1 \tag{32}$$

387 It proves that :

$$\begin{aligned}
& \sum_{i=1}^N L_i \times r_i + \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \\
& \leq \alpha_1 + \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \\
& \leq \sum_{i=1}^N R_i \times I_i + R_{N+1} \times O - \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O + \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \\
& \leq \sum_{i=1}^N R_i \times I_i + R_{N+1} \times O
\end{aligned} \tag{33}$$

388 As long as $\prod_{i=1}^N r_i \prod_{i=1}^N \hat{R}_i \leq \prod_{i=1}^{N+1} R_i$, inequality 29 holds using 33.

$$\begin{aligned}
& \prod_{i=1}^N r_i \prod_{i=1}^{N+1} \hat{R}_i + \sum_{i=1}^N L_i \times r_i + \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \\
& \leq \prod_{i=1}^{N+1} R_i + \sum_{i=1}^N L_i \times r_i + \sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \\
& \leq \prod_{i=1}^{N+1} R_i + \sum_{i=1}^N R_i \times I_i + R_{N+1} \times O
\end{aligned} \tag{34}$$

389 Therefore, with ideal ranks, $\forall i \ \hat{R}_i \leq R_i$ and $\sum_{i=0}^N r_i L_i \leq \alpha_1$, inequality 29 holds.

390 Assume $\forall i \ \prod_{i=1}^N r_i \prod_{i=1}^N \hat{R}_i > \prod_{i=1}^{N+1} R_i$, then consider α_2 as:

$$\alpha_2 = \frac{\prod_{i=1}^{N+1} R_i}{\prod_{i=1}^{N+1} \hat{R}_i} \tag{35}$$

391 As long as $\prod_{i=1}^N r_i \leq \alpha_2$, inequality 29 holds using 33. However, if $\prod_{i=1}^N r_i > \alpha_2$, then consider k
392 as:

$$k = \prod_{i=1}^N r_i \prod_{i=1}^N \hat{R}_i - \prod_{i=1}^{N+1} R_i \tag{36}$$

393 Given:

$$k \leq \left(\sum_{i=1}^N R_i \times I_i + R_{N+1} \times O \right) - \left(\sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O \right) \tag{37}$$

394 Therefore, with ideal ranks and splits, inequality 29 holds using 37. □

395 **semi ideal.** We prove that given semi ideal ranks, inequality 29 holds.

396 Assume $\forall i \ \hat{R}_i > R_i$, which is equivalent to $\exists j \ \hat{R}_j > R_j$, then:

$$k_i = \hat{R}_i - R_i \quad (38)$$

397 We can consider:

$$\sum_{i=1}^N \hat{R}_i \times \frac{I_i}{L_i} + \hat{R}_{N+1} \times O = \sum_{i=1}^N R_i \times \frac{I_i}{L_i} + \sum_{i=1}^N k_i \times \frac{I_i}{L_i} + R_{N+1} \times O + k_{N+1} \times O \quad (39)$$

398 Similar to TCL, consider β as:

$$\begin{aligned} \beta &= \sum_{i=1}^N R_i \times I_i - \sum_{i=1}^N R_i \times \frac{I_i}{L_i} \\ &= \sum_{i=1}^N R_i \times I_i \times \left(1 - \frac{1}{L_i}\right) \end{aligned} \quad (40)$$

399 Then as long as the k_i parts of equation 39 is less than β , the inequality 30 holds.

$$\sum_{i=1}^N k_i \times \frac{I_i}{L_i} + k_{N+1} \times O \leq \beta \quad (41)$$

400 Therefore, given the assumption and the inequality 41, with ideal splits, 39 holds. The rest of the
401 proof is similar to *ideal* situation. $\square$

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